

Alpha Asymmetry in Foreign Exchange Markets

An Investigation of Exploitability

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Abstract

This paper investigates whether distributional asymmetries in foreign exchange alpha signals represent exploitable market inefficiencies. Using EUR/JPY data sampled from November 2015–August 2025 (504 analysis-ready weekly observations from January 2016 onward), we find that the asymmetry premise itself largely dissolves under dependence-robust measurement: of five alpha signal types, only the volatility-expansion (coverage) signal exhibits skewness that survives block-bootstrap inference ($\hat{\gamma}_1 = 1.75$, 95% CI [1.18, 2.16]); the signed tail signal skews *negative* rather than positive (-1.48 , CI $[-3.10, 0.54]$), and momentum, mean-reversion, and correlation signals have skewness intervals that include zero. An earlier version of this paper reported pronounced positive tail-signal skewness; that figure described the unsigned exceedance magnitude, which is right-skewed by construction, and we document the correction. The economic null is correspondingly stark. Under the corrected chronology and holding rule, a skewness-threshold strategy loses 7.57% cumulative gross over the decade (15 directional holding episodes, Sharpe -0.17), is nearly inert in walk-forward testing (one episode in eight out-of-sample years), and provides no evidence of a positive residual alpha against proxy carry, momentum, and dollar factors (intercept $t = -0.86$ on in-position weeks). Transaction costs modestly worsen an already negative gross result; a positive-cost break-even level is therefore not defined. Data-snooping corrections complete the null: White's Reality Check ($p = 0.15$) and Hansen's SPA ($p = 0.26$) find no strategy in a 13-candidate universe that outperforms the zero-return benchmark—the best realized mean return belongs to a seeded random candidate. The weekly-return GPD shape parameter is imprecisely estimated ($\xi = -0.25$), and tail clustering is weak ($\theta = 0.83$, 95% CI $[0.60, 1.00]$). We conclude that the evidence does not support broad alpha signal asymmetry in this setting (with one distributional exception that does not yield a tradable edge) or its exploitability, and we caution that unsigned-magnitude constructions can manufacture an appearance of asymmetry that signed-data analysis does not support.

Keywords: null result, alpha asymmetry, foreign exchange, skewness, market efficiency, extreme value theory

JEL Codes: G11, G14, G15, C58

1 Introduction

Financial markets exhibit persistent deviations from the efficient market hypothesis (Fama, 1970), with alpha signals (measures of risk-adjusted excess returns) displaying systematic patterns that sophisticated traders exploit (Jegadeesh and Titman, 1993; Moskowitz et al., 2012). While considerable literature examines alpha generation and decay, less attention has been paid to the *distributional properties* of alpha signals themselves. This is surprising given well-documented evidence that asset returns deviate substantially from normality, exhibiting fat tails (Mandelbrot, 1963; Cont, 2001) and asymmetric distributions (Harvey and Siddique, 2000).

The preference for skewed returns has deep roots in asset pricing theory. Kraus and Litzenberger (1976) established that investors prefer positive skewness and dislike negative skewness, implying that

assets with lottery-like payoffs command lower expected returns. Subsequent work has confirmed that skewness affects both individual asset pricing (Harvey and Siddique, 2000; Bali et al., 2011) and portfolio construction (Mitton and Vorkink, 2007; Brunnermeier et al., 2007).

This paper investigates whether alpha signals exhibit systematic asymmetries in their probability distributions, and whether such asymmetries can inform profitable trading strategies. We report negative findings that begin earlier in the chain than the usual backtest disappointment: under dependence-robust measurement, most of the proposed asymmetry is not supported. We investigate the following questions:

1. Do alpha signals in forex markets deviate significantly from normal distributions? (*Mostly: four of five types reject normality; normality is not rejected for the normalized-return signal*)
2. Do these deviations manifest as exploitable skewness and heavy tails? (*Largely no: one signal is robustly skewed, the tail signal skews negative and fragilely, and the tail-shape estimate is too imprecise to establish Pareto-type heavy tails*)
3. Do distributional asymmetries persist across market regimes? (*No: the tail signal's skewness flips sign across regimes*)
4. Do asymmetry-aware strategies outperform benchmarks? (*No: returns are negative before costs and trail buy-and-hold outright, the strategy is dormant under walk-forward selection, and no candidate significantly beats the zero-return benchmark*)

We document a plausible-sounding trading idea that does not survive rigorous testing, and we document *where* it fails: earlier than expected. The asymmetry-exploitation thesis does not fall at the translation from statistical pattern to economic profit; it falls at signal detection, where an unsigned-magnitude construction had manufactured the appearance of pervasive positive skewness that corrected measurement does not support. Null results of this kind are underreported in quantitative finance (Harvey, 2017), yet they prevent wasted research effort and capital allocation to spurious patterns.

The FX factor literature has seen recent advances in factor construction methodology. Fan et al. (2025) introduce a framework that dynamically optimizes currency factor strategies, including carry, momentum, and value, via spot and forward trading, evaluating 24,336 portfolio optimization approaches and demonstrating that optimized factors significantly outperform naïve constructions after correcting for data snooping bias. Our work complements this optimization-focused agenda by examining a more fundamental question: whether the *distributional asymmetry* in factor returns (long vs. short side) itself constitutes an exploitable signal. Where Fan et al. optimize how factors are constructed, we test whether the asymmetric properties of the resulting return distributions carry economic information. Hertrich (2025) provides additional context through analysis of G10 carry, momentum, and value strategies under forward-looking Conditional Value-at-Risk conditioning, finding that the carry trade risk premium has remained unexpectedly low since the global financial crisis despite negative covariance with global FX volatility—suggesting a potential failure of currency pricing theory that our distributional analysis may help illuminate.

Our empirical contribution is threefold. First, we provide a systematic characterization of distributional properties across five distinct alpha types in EUR/JPY forex data. Second, the corrected strategy loses money before frictions, and walk-forward selection leaves it nearly dormant. Third, unchanged cross-market transfers range from a 14.11% loss to a 13.18% gain, without market-specific costs or inference; these are exploratory instability checks, not evidence of generalization.

2 Methodology

2.1 Data and Sample Construction

We sample EUR/JPY forex data from November 2015 to August 2025, sourced from Yahoo Finance (EURJPY=X). The raw weekly grid begins on 6 November 2015; after rolling-statistic warmup, the analysis-ready sample runs from 8 January 2016 to 29 August 2025 and contains 504 weekly observations. We note that Yahoo Finance quotes are indicative mid-rates rather than executable prices; accordingly, all transaction cost analysis (Section 5.4) uses separate spread assumptions. The repository did not contain the raw snapshots used for the previously committed results, so this revision records retrieval times and SHA-256 hashes but does not claim bit-for-bit reproduction of that earlier run. The sample period encompasses multiple market regimes including the post-Brexit volatility spike (2016), COVID-19 pandemic shock (2020), and subsequent monetary policy divergence between the Federal Reserve, ECB, and Bank of Japan.

Data Frequency and Aggregation. Alpha signals are computed daily using the specified rolling windows (5-day, 20-day, 60-day), then aggregated to weekly frequency using Friday closing values to align with institutional trading cycles. The 504 analysis-ready weekly observations represent end-of-week snapshots of daily-computed signals. Daily returns feed the alpha definitions and volatility estimates; strategy returns are Friday-close-to-Friday-close percentage changes.

The dataset includes five pre-computed alpha signals, each capturing distinct aspects of market dynamics:

- **Tail Alpha (α_{tail}):** Captures extreme daily price movements without discarding their direction. Formally, $\alpha_{\text{tail},t} = \mathbf{1}_{|r_t| > q_{0.95,t}} \cdot r_t$, where $q_{0.95,t}$ denotes the 95th percentile of absolute daily returns over the trailing 252 trading days (with 60 observations required during warm-up). The Friday observation is retained in the weekly signal panel.
- **Fast Alpha (α_{fast}):** Short-term momentum signal computed as the 5-day percentage return normalized by 20-day realized volatility: $\alpha_{\text{fast},t} = (P_t/P_{t-5} - 1)/(\sigma_{20,t}\sqrt{5})$.
- **Pricing Alpha (α_{price}):** Mean-reversion signal measuring deviation from fair value, computed as $\alpha_{\text{price},t} = (P_t - \text{MA}_{60,t})/\sigma_{60,t}$, where MA_{60} and σ_{60} denote 60-day moving average and standard deviation.
- **Coverage Alpha (α_{cov}):** Volatility compression ratio measuring regime shifts in realized volatility:

$$\alpha_{\text{cov},t} = \frac{\sigma_{20,t}}{\sigma_{20,t-5}} - 1 \quad (1)$$

where $\sigma_{20,t}$ denotes 20-day realized volatility computed from daily close-to-close returns. Data source: Yahoo Finance (EURJPY=X). Values above zero indicate volatility expansion; values below zero indicate compression.

- **Hedge Alpha (α_{hedge}):** A fixed-rate correlation proxy, defined as:

$$\alpha_{\text{hedge},t} = \rho_t^{(\text{EUR/JPY,DXY})} \times (-0.02) \quad (2)$$

where ρ is a 100-trading-day (≈ 20 -week) rolling correlation of daily EUR/JPY and DXY returns (US Dollar Index, DX-Y.NYB from Yahoo Finance). The code does not use a time-varying

interest-rate series: it multiplies the correlation by a fixed -2% constant. Hedge alpha is therefore a fixed-rate correlation proxy, not a measure of changing interest-rate regimes. Replacing the constant with a dated EUR–JPY rate differential or forward-implied measure is left for future work.

For cross-market validation, we apply the same transformations to GBP/USD, SPY (S&P 500 ETF), and GLD (Gold ETF) daily closes from Yahoo Finance, constructing tail, fast, pricing, and coverage alphas identically to EUR/JPY. Hedge alpha is excluded from the cross-market set because the DXY-correlation proxy was specified for the EUR/JPY setting and is not a genuine transferable rate-differential signal.

2.2 Asymmetry Metrics

We employ four complementary measures to characterize distributional asymmetry, each capturing distinct aspects of non-normality relevant to trading strategy design.

Sample Skewness measures the degree of asymmetry around the mean. For a sample $\{x_1, \dots, x_n\}$ with sample mean $\bar{x}$ and standard deviation s , we compute the adjusted Fisher-Pearson coefficient:

$$\hat{\gamma}_1 = \frac{n}{(n-1)(n-2)} \sum_{i=1}^n \left(\frac{x_i - \bar{x}}{s} \right)^3 \quad (3)$$

This bias-corrected estimator is consistent under standard regularity conditions. Positive skewness ($\hat{\gamma}_1 > 0$) indicates a right-tailed distribution with more extreme positive values, while negative skewness indicates left-tail heaviness. The standard error of skewness under normality is approximately $\sqrt{6/n} \approx 0.109$ for our sample size ($n = 504$).

Excess Kurtosis captures tail heaviness relative to a Gaussian benchmark:

$$\hat{\gamma}_2 = \frac{n(n+1)}{(n-1)(n-2)(n-3)} \sum_{i=1}^n \left(\frac{x_i - \bar{x}}{s} \right)^4 - \frac{3(n-1)^2}{(n-2)(n-3)} \quad (4)$$

This estimator subtracts 3 (the kurtosis of a normal distribution) and applies finite-sample bias correction. Values exceeding zero indicate leptokurtic (fat-tailed) distributions; the standard error under normality is approximately $\sqrt{24/n} \approx 0.218$.

Asymmetry Index (AI) quantifies the ratio of upside to downside semi-variance, providing a risk-management-oriented asymmetry measure:

$$AI = \frac{\text{Var}^+(X)}{\text{Var}^-(X)} = \frac{\sum_{i: x_i > \bar{x}} (x_i - \bar{x})^2 / n^+}{\sum_{i: x_i < \bar{x}} (x_i - \bar{x})^2 / n^-} \quad (5)$$

where n^+ and n^- denote observations above and below the mean. Values above 1 indicate greater dispersion in positive deviations—relevant for assessing “lottery ticket” payoff structures (Bali et al., 2011). The index is undefined when either side is empty or the downside mean-squared deviation is zero; the implementation returns a missing value in those cases. A single observation on one side remains valid because the calculation is a mean square about the overall mean, not a subgroup sample variance.

Positive Observation Ratio (POR) measures the unconditional probability of positive realizations:

$$POR = \frac{1}{n} \sum_{i=1}^n \mathbf{1}_{x_i > 0} \quad (6)$$

Under normality with zero mean, $POR = 0.5$; deviations indicate location-scale asymmetry or non-zero drift.

2.3 Statistical Tests

We employ a battery of complementary tests to assess departures from normality, each with distinct power properties against different alternatives.

Shapiro-Wilk Test. The Shapiro-Wilk statistic (Shapiro and Wilk, 1965) tests the null hypothesis $H_0 : X \sim \mathcal{N}(\mu, \sigma^2)$ against general alternatives. The test statistic is:

$$W = \frac{(\sum_{i=1}^n a_i x_{(i)})^2}{\sum_{i=1}^n (x_i - \bar{x})^2} \quad (7)$$

where $x_{(i)}$ are order statistics and a_i are tabulated coefficients. This test has high power against asymmetric and heavy-tailed alternatives for moderate sample sizes.

D'Agostino-Pearson Omnibus Test. Following D'Agostino et al. (1990), we test skewness and kurtosis jointly using the K^2 statistic:

$$K^2 = Z_1(\hat{\gamma}_1)^2 + Z_2(\hat{\gamma}_2)^2 \sim \chi_2^2 \quad (8)$$

where Z_1 and Z_2 are normalizing transformations of sample skewness and kurtosis. This test is particularly powerful against asymmetric alternatives.

Jarque-Bera Test. As a robustness check, we employ the Jarque-Bera statistic (Jarque and Bera, 1980):

$$JB = \frac{n}{6} \left(\hat{\gamma}_1^2 + \frac{(\hat{\gamma}_2)^2}{4} \right) \stackrel{a}{\sim} \chi_2^2 \quad (9)$$

This Lagrange multiplier test is asymptotically equivalent to the likelihood ratio test for normality.

Skewness Significance. We test $H_0 : \gamma_1 = 0$ using the t -ratio $t = \hat{\gamma}_1 / \text{SE}(\hat{\gamma}_1)$, where $\text{SE}(\hat{\gamma}_1) \approx \sqrt{6(n-2)/[(n+1)(n+3)]}$. For $n = 504$, this yields a standard error of approximately 0.109.

Multiple Testing Correction. Given five alpha types and multiple test statistics, we apply the Bonferroni correction to control the family-wise error rate at $\alpha = 0.05$, yielding adjusted significance thresholds of $\alpha^* = 0.05/5 = 0.01$ per alpha type.

2.4 Backtesting Framework

We evaluate the asymmetry strategy against three benchmarks, with explicit signal generation and execution rules to ensure reproducibility.

Asymmetry Strategy:

Signal Generation:

- **Entry Long:** Rolling skewness of fast alpha (20-week window) exceeds 0.75 AND current fast alpha > 0
- **Entry Short:** Rolling skewness of pricing alpha (20-week window) exceeds 0.75 AND current pricing alpha $> 0.5\sigma_{\text{pricing}}$
- **Exit or reversal:** Preserve an open position through no-signal and same-direction-signal weeks; reverse directly when the opposing entry condition is met; otherwise exit after four realized weekly return periods. An opposing signal takes precedence if it arrives on the expiry date.

- **Simultaneous signals:** Close any open position and remain flat; do not open a new position.

Position Sizing:

$$\text{Position size} = \min(2.0, 1 + |AI_t - 1.0|) \quad (10)$$

where AI_t is the contemporaneous asymmetry index. Because $AI_t \geq 0$, the formula spans 1–2 gross-notional units; the previously stated 0.5-unit lower bound was unreachable. A missing AI uses the neutral one-unit size. Size is fixed at entry and is not reset or resized by subsequent same-direction signals.

Execution:

- Signal and price convention: signals are observed at each Friday close and the same Friday close is used as an indicative execution proxy
- Return convention: that position earns only the subsequent Friday-close-to-Friday-close return (one lag)
- Rebalancing: none within an episode; changes occur only on entry, reversal, conflict, or expiry
- Exposure: positions are bounded to $[1.0, 2.0]$ gross-notional units; values above one imply leverage relative to one unit of capital

Monday opening prices are not present in the dataset, so the backtest does not claim Monday-open execution. The Friday-close proxy is an internally consistent bar-to-bar convention, not evidence that the displayed mid-price could have been filled after observing that exact close. All headline, benchmark, walk-forward, factor, cost, and data-snooping returns use the same single-lag convention.

Momentum Benchmark: Weekly time-series momentum (Moskowitz et al., 2012): long one unit when the trailing 20-week return is positive, short one unit when it is negative, so the position is the sign of the trailing return and the strategy is always in the market once the lookback window fills.

Mean Reversion Benchmark: Contrarian positions on the z -score of the weekly close against its 20-week moving average: short one unit when $z_t > 2$, long one unit when $z_t < -2$, flat otherwise (positions close when $|z|$ falls back inside the two-standard-deviation band).

Buy & Hold: A constant long position of one unit, entered at the start of the sample and held throughout; its cumulative return equals the EUR/JPY spot move over the sample.

The momentum and mean-reversion benchmarks are the same rules that enter the data-snooping universe, evaluated with positions applied to the following week’s return. Performance metrics include cumulative return, volatility, Sharpe and Sortino ratios, drawdown, directional holding episodes, execution legs, reversals, resizing, and turnover.

3 Results

3.1 Asymmetry Detection

Table 1 presents distributional statistics for each alpha type.

Table 1: Alpha Asymmetry Metrics

Alpha Type	Skew	Kurt	AI	POR
Tail	-1.48	19.20	0.03	2.78%
Fast	0.01	0.01	0.98	51.98%
Pricing	-0.17	-0.72	0.77	57.54%
Coverage	1.75	7.65	2.22	53.17%
Hedge	0.15	-0.51	1.01	53.57%

Note: Skew = bias-corrected sample skewness, Kurt = bias-corrected excess kurtosis, AI = ratio of mean squared positive and negative deviations from the overall mean, POR = positive observation ratio. All statistics use the signed weekly series ($n = 504$). The earlier AI code used subgroup sample variances, which recentred each side and used $n - 1$ denominators; the values above implement the displayed equation.

The corrected statistics tell a very different story from the one asymmetry-based trading would require. The single robust asymmetry is in *coverage* alpha, the volatility-expansion signal, which is strongly right-skewed (1.75) with high excess kurtosis (7.65) and an asymmetry index of 2.22: volatility expands abruptly and contracts gradually. This is a familiar feature of volatility dynamics, not a tradable anomaly in itself. Tail alpha, by contrast, has a negative point estimate (-1.48), although its dependence-robust interval includes zero. Its POR of 2.78% reflects that the series is zero except at exceedances. Normality is not rejected for fast alpha; this does not establish that its population distribution is exactly Gaussian. Statistical asymmetry, where detected, is distinct from profitable predictability.

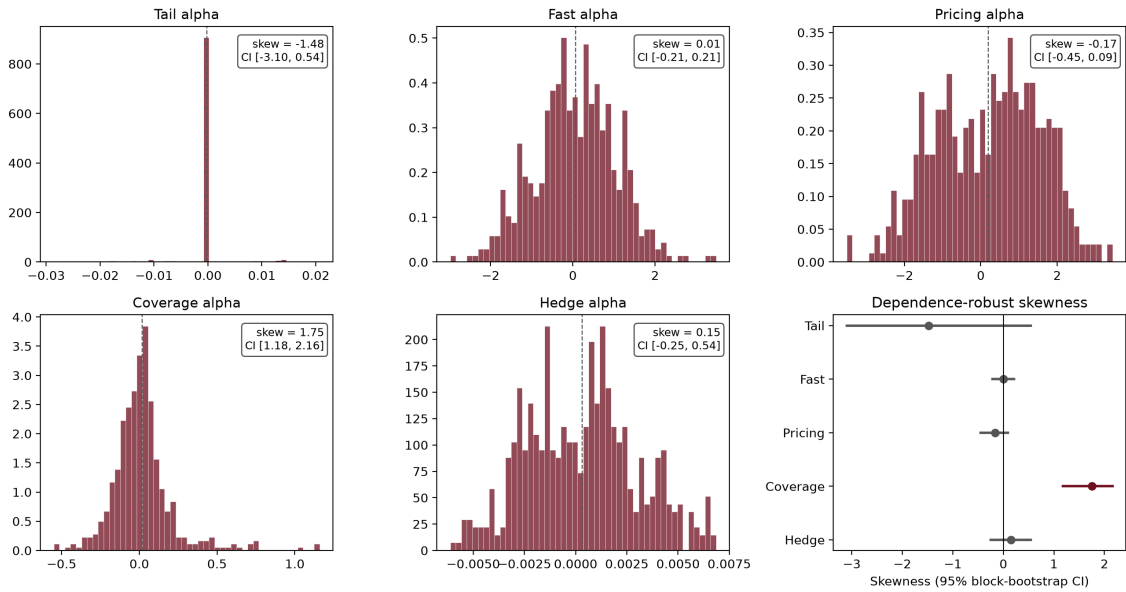
Signed weekly alpha signals, EUR/JPY (2016-01-08 to 2025-08-29, $n=504$)

Figure 1: Alpha Asymmetry Analysis. Distributions of the five signed weekly alpha signals (EUR/JPY analysis-ready sample, January 2016–August 2025, $n = 504$) with skewness point estimates and 95% circular block bootstrap confidence intervals; the final panel summarizes the dependence-robust skewness inference. Only coverage alpha’s interval excludes zero.

3.2 Statistical Significance

Table 2 presents hypothesis test results for each alpha type. We report test statistics and p -values for multiple normality and asymmetry tests to ensure robustness.

Table 2: Statistical Test Results

Alpha	$\hat{\gamma}_1$	t -stat	Block CI	SW	JB	K^2	LB $Q(4)$
Tail	-1.48	-13.5***	[-3.10, 0.54]	0.335***	7922***	244.4***	12.6*
Fast	0.01	0.1	[-0.21, 0.21]	0.997	0.0	0.0	6.5
Pricing	-0.17	-1.5	[-0.45, 0.09]	0.984***	13.3**	30.1***	679.1***
Coverage	1.75	16.1***	[1.18, 2.16]	0.865***	1489***	217.6***	72.6***
Hedge	0.15	1.4	[-0.25, 0.54]	0.990**	7.4*	11.7**	1716.2***

Note: t -stat tests $H_0 : \gamma_1 = 0$ with $SE = \sqrt{6/n} = 0.109$ ($n = 504$) and assumes i.i.d. observations; Block CI = 95% circular block bootstrap confidence interval for $\hat{\gamma}_1$ (block length 13 weeks, $B = 2000$), which is robust to the serial dependence the i.i.d. tests ignore; SW = Shapiro-Wilk statistic; JB = Jarque-Bera statistic; K^2 = D’Agostino-Pearson omnibus; LB $Q(4)$ = Ljung-Box statistic at four lags, reported per series. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$ (unadjusted); the qualitative conclusions below apply Bonferroni correction at family size $m = 5$. Because pricing, coverage, and hedge alphas are built from overlapping rolling windows, their Ljung-Box statistics are large by construction and the block-bootstrap intervals are the appropriate basis for skewness inference.

Normality. Four of five alpha types reject normality. For fast alpha, none of the reported tests rejects the Gaussian null (SW $p = 0.55$, JB ≈ 0 , K^2 $p = 0.99$). This is compatibility with the null in this sample, not proof of exact Gaussianity. Tail and coverage depart severely; pricing and hedge reject more mildly.

Skewness. Under the i.i.d. t -test, tail and coverage alphas show large-magnitude skewness of opposite signs. But the alphas are serially dependent; pricing, coverage, and hedge have Ljung-Box statistics ranging from 72.6 to 1716, so the block-bootstrap intervals are the appropriate inferential yardstick. On that yardstick only *coverage* alpha survives: its interval [1.18, 2.16] excludes zero decisively, while the tail-alpha interval [-3.10, 0.54] widens to include zero once dependence is respected (the point estimate is dominated by a small number of clustered exceedance weeks). Fast, pricing, and hedge alpha intervals include zero.

What survives. The asymmetry inventory that motivates the rest of the paper is therefore modest: one robustly right-skewed volatility-expansion signal, one negatively-skewed-but-fragile tail signal, and three signals whose skewness intervals include zero. This is consistent with the broader stylized facts on asset-price distributions (Cont, 2001; Mandelbrot, 1963) while falling well short of the pervasive “alpha asymmetry” a trading strategy would want.

3.3 Backtest Performance

Table 3 compares the asymmetry strategy against the three benchmarks of Section 2.4 on the EUR/JPY analysis-ready sample (January 2016–August 2025). We report standard performance measures alongside risk-adjusted metrics to enable fair comparison across strategies with different return and volatility profiles.

Table 3: Strategy Performance Comparison (EUR/JPY analysis-ready sample, January 2016–August 2025)

Strategy	Return	Vol	Sharpe	Sortino	MDD	Episodes / Legs
Asymmetry	−7.57%	4.20%	−0.173	−0.090	−14.29%	15 / 30
Momentum (20w)	−18.92%	8.40%	−0.215	−0.309	−31.32%	54 / 107
Mean Rev. (2.0σ)	10.74%	3.02%	0.364	0.198	−4.32%	27 / 54
Buy & Hold	33.53%	8.60%	0.390	0.584	−15.78%	1 / 1

Note: Return = cumulative gross return over 504 weeks; Vol = annualized volatility; MDD = maximum drawdown. A holding episode is one continuous directional position; a direct reversal closes one episode and opens another. Execution legs count entries and exits separately, so a reversal has two legs. Turnover, resizing, and reversals are also recorded in the dated ledgers. All strategies use one timing lag.

Risk-Adjusted Performance. The corrected asymmetry strategy loses 7.57% gross, with a Sharpe ratio of -0.173 , versus a 33.53% buy-and-hold gain. It is invested for 55 of 504 realized-return weeks. Its low 4.20% annualized volatility still largely reflects inactivity rather than skill. The 20-week momentum benchmark also loses money, while the simple mean-reversion and buy-and-hold benchmarks are positive.

Drawdown Analysis. The asymmetry strategy’s maximum drawdown is -14.29% , only modestly smaller than buy-and-hold’s -15.78% despite far lower exposure. Mean reversion remains more contained at -4.32% .

Trading Frequency. The corrected ledger records 15 asymmetry holding episodes, 30 execution legs, one direct reversal, no resizing, and total absolute turnover of 49.15 units. Momentum has 54 directional episodes because every sign reversal closes one episode and opens another; dividing position-change events by two understated this count. These definitions separate economic holding episodes from executions and turnover.

Statistical Significance of Returns. Stationary-bootstrap inference (expected block length 4 weeks, $B = 2000$) gives an annualized return of -0.81% with a 95% interval of $[-3.57\%, 1.48\%]$ and a Sharpe interval of $[-0.76, 0.41]$. The sample estimate is negative and the interval remains too wide to establish a nonzero population mean. Failure to reject zero is not proof that the true effect is exactly absent.

Strategy equity curves and drawdowns, one-lag Friday-close convention

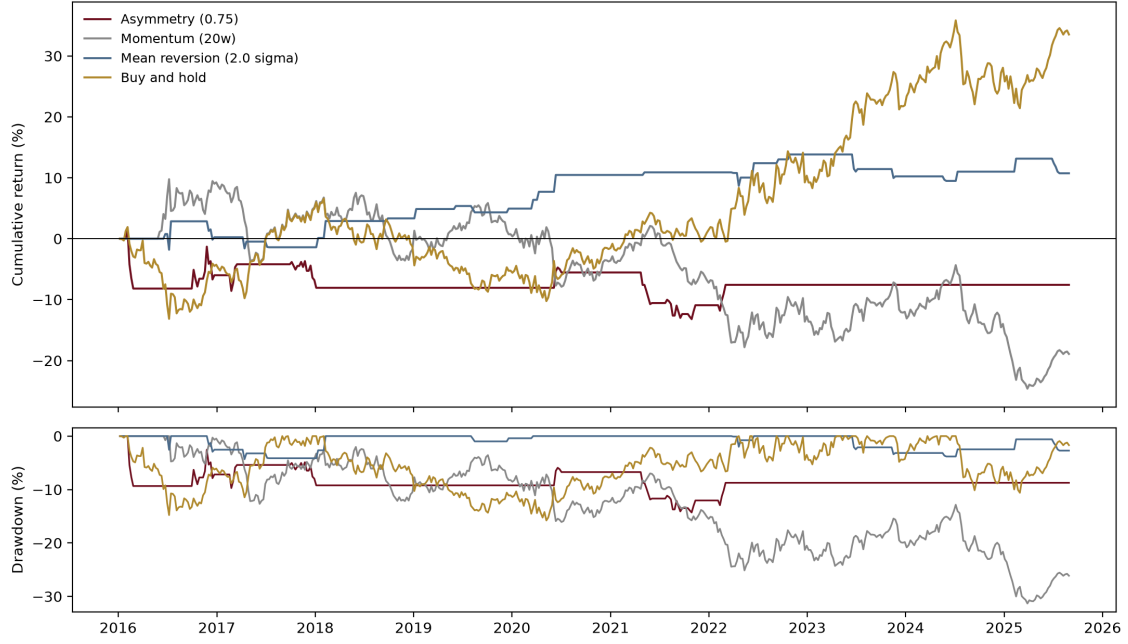


Figure 2: Strategy equity curves and drawdowns. Cumulative gross returns (top) and drawdowns (bottom) under the common one-lag convention. Generated by `analysis/full_pipeline.py`.

3.4 Cross-Market Validation

To test whether the EUR/JPY findings generalize, we apply the identical alpha construction and the identical trading strategy (unchanged code, unchanged parameters) to GBP/USD, SPY, and GLD (Section 2.1). Hedge alpha is excluded as EUR/JPY-specific; the four remaining signals and the two-sided asymmetry strategy (0.75 threshold) transfer directly. Table 4 reports the signed weekly skewness of each signal alongside the strategy’s gross cumulative return and, for scale, each market’s buy-and-hold return over the same 504-week window.

Table 4: Cross-Market Check

Market	Tail Skew	Fast Skew	Pricing Skew	Coverage Skew	Strat. Ret.	B&H Ret.
EUR/JPY	−1.48	0.01	−0.17	1.75	−7.57%	33.53%
GBP/USD	0.82	0.01	−0.14	1.58	−14.11%	−7.63%
SPY	0.94	−0.21	−1.09	1.85	13.18%	294.30%
GLD	0.69	−0.00	−0.19	1.32	−13.67%	200.97%

Note: Bias-corrected skewness of signed weekly signals ($n = 504$ per market). Strategy returns use the corrected one-lag strategy with 15, 19, 4, and 17 directional holding episodes respectively. SPY uses adjusted prices, explicitly requested from the data provider. Point estimates do not include market-specific block-bootstrap inference or costs; this remains exploratory.

The distributional signature does not travel: EUR/JPY’s negative tail-skew point estimate is positive in all three other markets, while coverage alpha’s right skew recurs. Strategy outcomes are likewise unstable: the unchanged rules lose 14.11% on GBP/USD, gain 13.18% on SPY while trailing buy-and-hold by a wide margin, and lose 13.67% on GLD. These exploratory transfers lack pair-specific costs and dependence-aware inference, so they do not establish effects in any market.

Scope Limitation. This analysis focuses on EUR/JPY as a liquid, widely-traded cross rate with

substantial institutional participation. Cross-market results (GBP/USD, SPY, GLD) are presented as an illustrative transfer check, not a definitive multi-asset analysis. A comprehensive treatment would require: (i) asset-specific alpha calibration reflecting different market microstructures; (ii) market-specific transaction cost modeling (FX spreads vs. equity commissions vs. futures margins); and (iii) asset-by-asset statistical inference with proper multiple-testing corrections. We leave this extension to future work; on the evidence here, nothing suggests the EUR/JPY null would reverse elsewhere.

4 Discussion

4.1 Why Asymmetry Exploitation Fails

The empirical findings reveal a pattern more thoroughgoing than “asymmetry exists statistically but fails economically”: under corrected measurement, most of the proposed asymmetry lacks dependence-robust support, and the one robust distributional finding does not establish a tradable edge. We identify four failure modes:

Little Asymmetry to Exploit. The dependence-robust inventory leaves one robustly skewed signal (coverage) and one fragile negative point estimate (tail). The trading rule instead uses fast and pricing skewness, whose full-sample intervals include zero. Its 15 in-sample episodes do not establish profitable predictability.

Tail Evidence Is Inconclusive. The GPD point estimate is near zero, but its wide confidence interval includes both negative and positive values. The data do not establish Pareto-type heavy tails, nor do they prove exponential decay. With only 20 declustered maxima, tail classification is underpowered.

Inactivity Under Honest Selection. When the threshold is chosen on training data alone, the strategy opens only one OOS holding episode in eight test years. That count is too small for performance inference.

Non-Generalizability. Cross-market transfer yields erratic results: the identical rules lose 14.11% on GBP/USD, gain 13.18% on SPY while trailing buy-and-hold by roughly 281 percentage points, and lose 13.67% on GLD. The tail signal’s negative point estimate also reverses sign in every other market tested. Without asset-specific costs or inference, nothing here establishes generalization.

Transaction costs are not the cause of the failure: the corrected gross return is already negative, and costs make it monotonically worse (Section 5.4).

4.2 What the Asymmetry Statistics Do Show

The corrected statistics still reveal genuine features of market structure, just not the ones an asymmetry trader would want:

Volatility-Expansion Asymmetry: Coverage alpha’s robust right skew (1.75, block CI [1.18, 2.16]) is consistent with a familiar asymmetry in volatility dynamics— σ can spike abruptly and decay more slowly (Bollerslev, 1986; Engle, 1982). Nothing in our results shows that this distributional feature predicts the direction of spot returns.

Downside-Leaning Tails: The signed tail signal has a negative skewness point estimate, indicating that the observed extreme negative magnitudes dominate the positive tail in this sample. That direction is the opposite of a lottery-ticket structure (Kumar, 2009; Bali et al., 2011) and closer to the crash-risk asymmetry of the carry literature (Brunnermeier et al., 2009; Burnside et al., 2011). The dependence-robust interval $[-3.10, 0.54]$ includes zero, so the estimate is too fragile to support a strategy.

A Near-Symmetric Baseline: The tests do not reject normality for fast alpha in this sample. That result is consistent with a near-symmetric normalized-return signal, but it does not prove exact Gaussianity or rule out other predictive structure (Moskowitz et al., 2012).

4.3 Limitations

Several limitations warrant acknowledgment. First, 504 weekly observations and only 15 holding episodes provide limited power, especially for tail and trade-level inference. Second, the alpha definitions are fully disclosed here but are ad hoc and may not generalize. Third, the factor exercise uses spot proxies rather than published cross-sectional FX factors; even with 55 in-position weeks, the effective sample is small, and the significant negative momentum loading is descriptive rather than a stable structural estimate. Fourth, costs use stylized spreads and indicative mid-prices, omit financing, and assume end-of-bar execution; no Monday-open or executable bid/ask data are available. Fifth, cross-market transfer omits pair-specific financing, costs, and dependence-aware inference. Finally, Yahoo history can be revised. The original raw snapshots were not committed, so exact reproduction of the archived figures is impossible without the owner’s files; the revised pipeline caches local inputs and records hashes and software versions.

5 Robustness Analysis

The preceding sections leave a thin asymmetry inventory (one robustly skewed signal among five) and an in-sample backtest whose returns are statistically indistinguishable from zero. This section stress-tests what remains, addressing four questions: (1) Do results hold out-of-sample? (2) Are patterns stable across market regimes? (3) Do trading costs erode strategy returns? (4) Is asymmetry alpha distinct from known FX risk factors?

5.1 Out-of-Sample Validation

A primary concern with any trading strategy is in-sample overfitting. Our main results use the full January 2016–August 2025 analysis-ready sample, which may inadvertently optimize signal construction to historical patterns that do not persist. We address this through walk-forward analysis, a methodology widely used in quantitative finance to assess genuine predictive power (Lo et al., 2000).

We implement an expanding-window walk-forward: the entry threshold is selected from the grid $\{0.50, 0.75, 1.00, 1.25\}$ by Sharpe ratio on all analysis-ready data through year $t - 1$ (beginning with 2016–2017), then the strategy runs untouched on calendar year t , rolling forward annually through 2025. All signal construction uses trailing windows only, so no test-year information enters signal values; the walk-forward isolates the threshold-selection step, which is the strategy’s only free parameter. Table 5 presents out-of-sample performance metrics.

Table 5: Out-of-Sample Walk-Forward Results

Test Year	Selected Threshold	OOS Return	OOS Sharpe	Hit Rate	Episodes
2018	1.25	0.00%	0.00	—	0
2019	1.25	0.00%	0.00	—	0
2020	1.25	2.74%	0.88	50.0%	1
2021	1.25	0.00%	0.00	—	0
2022	1.25	0.00%	0.00	—	0
2023	1.25	0.00%	0.00	—	0
2024	1.25	0.00%	0.00	—	0
2025	1.25	0.00%	0.00	—	0
Pooled OOS	—	2.74%	0.32	50.0%	1

Note: The threshold is selected using data through the prior year, then applied on one sequential OOS state path. Episodes are newly opened directional holdings, not position-change pairs. Pooled return is cumulative (0.35% annualized); 2025 ends in August. With one episode, these metrics are descriptive only.

The corrected training windows select 1.25 throughout. The strategy opens one OOS episode, in 2020, and remains flat in the other test years. Its 2.74% pooled gain therefore cannot be interpreted as evidence of skill: it is one realized path from one holding episode.

5.2 Subsample Stability

Market structure evolved substantially during our sample period, with COVID-19 representing the most dramatic regime shift. We test whether asymmetry patterns persist across distinct market environments using subsample analysis.

Table 6 partitions the sample into economically meaningful subperiods.

Table 6: Subsample Asymmetry Stability

Subsample	N	Tail Skew	Fast Skew	Price Skew	Strategy Ret.
Pre-COVID (2016–2019)	208	-3.40	-0.05	0.12	-8.06%
COVID Shock (2020)	52	5.02	0.09	-0.07	2.74%
Post-COVID (2021–2025)	244	-0.73	0.05	-0.38	-2.15%
Low VIX ($VIX < 20$)	342	-3.25	0.04	-0.20	-5.78%
High VIX ($VIX \geq 20$)	162	0.62	-0.06	-0.10	-1.90%
Rate Hike Regime (2022–2025)	191	-0.71	-0.13	-0.53	3.76%

Note: The strategy is run once on the complete chronological calendar; realized weekly returns are then attributed to rows in each regime. VIX is the mean of daily closes within each Friday-ending week. No stateful strategy is rerun on filtered, nonconsecutive dates. Skewness uses the observations in each displayed subsample.

Temporal Stability. Tail skewness is strongly negative pre-COVID (-3.40), positive in 2020 (+5.02), and moderately negative thereafter (-0.73). These point estimates are based on small, sparse exceedance samples and are descriptive. Fast skew remains near zero; pricing skew is more negative after 2020.

Regime Conditioning. Low- and high-VIX attribution is negative in both groups (-5.78% and -1.90%). Because these are ex post return attributions rather than separately tradable conditioned strategies, they describe where gains and losses occurred; they do not establish a usable VIX timing rule.

5.3 Entry Threshold Sensitivity

The baseline skewness threshold (0.75) deserves scrutiny: is it arbitrary or robustly selected? Table 7 reports strategy performance across alternative thresholds, holding other parameters constant.

Table 7: Sensitivity to Entry Threshold

Threshold	Return	Sharpe	MDD	Episodes	Hit Rate
0.50	−6.83%	−0.12	−22.02%	25	46.1%
0.75	−7.57%	−0.17	−14.29%	15	47.3%
1.00	−1.68%	−0.08	−5.46%	4	43.8%
1.25	2.74%	0.28	−0.86%	1	50.0%

Note: Returns use the full asymmetry strategy specification (long and short legs with position sizing as described in Section 2.4). Performance varies non-monotonically with threshold. Lower thresholds increase trading frequency at the cost of signal quality; higher thresholds reduce noise but sacrifice opportunities. Simplified long-only implementations yield different (typically lower) returns due to the absence of the short-side component.

Threshold Selection. The full-sample grid is non-monotonic. Three thresholds lose money; 1.25 earns 2.74% from a single episode. Selecting that isolated outcome after seeing the full sample would be data snooping, not robustness. The prespecified 0.75 baseline remains the confirmatory result, and the walk-forward separately freezes each year’s threshold from prior data.

5.4 Transaction Cost Sensitivity

Backtests without transaction costs overstate implementable returns. We analyze cost sensitivity across scenarios ranging from zero-cost (theoretical benchmark) to retail-level spreads.

The EUR/JPY market is highly liquid, with typical spreads of 0.5–1.0 pips during active trading hours (King et al., 2013). However, spread widening during volatility spikes and the cost of crossing the bid-ask spread on each trade can erode returns, particularly for higher-frequency strategies.

Table 8: Transaction Cost Analysis

Cost Scenario	Round-Trip Cost (pips)	Net Return	Net Sharpe
Zero Cost (Baseline)	0.0	−7.57%	−0.173
Prime Brokerage	0.3	−7.63%	−0.174
Institutional	0.7	−7.70%	−0.176
Retail (Tight)	1.3	−7.81%	−0.179
Retail (Wide)	2.0	−7.93%	−0.182

Note: Each execution leg is charged half the round-trip pips in proportion to absolute notional change at that row’s indicative price. Financing is excluded; because exposure can exceed one unit and lasts 55 weeks, its omission is a limitation, not assumed immaterial. A positive-cost break-even is not applicable because the zero-cost return is already negative.

Cost Components. We model spread and impact per dated execution leg. Financing/carry is excluded uniformly because no suitable dated rate or forward series is present; the omission may matter for leveraged positions and should be addressed with owner-approved data in future work.

Cost Interpretation. Execution costs make the result more negative. Gross return is −7.57%; at two pips round-trip, net return is −7.93%. The gross backtest fails first.

5.5 Factor Attribution

Does asymmetry alpha merely proxy for known FX risk factors? Prior work documents systematic risk premia in currency markets associated with carry (Lustig et al., 2011), momentum (Menkhoff et al.,

2012), and the dollar factor (Verdelhan, 2018), with Lettau et al. (2014) showing that downside risk conditioning explains a substantial portion of currency risk premia across asset classes. A parallel factor decomposition approach applied to cryptocurrency markets (Farzulla, 2026a) similarly finds that structural factors dominate idiosyncratic signal variation. If asymmetry signals simply load on these factors, our contribution reduces to factor timing rather than genuine alpha discovery.

We regress asymmetry strategy returns on proxy FX factors:

$$R_{\text{asym},t} = \alpha + \beta_1 \cdot \text{Carry}_t + \beta_2 \cdot \text{Mom}_t + \beta_3 \cdot \text{Dollar}_t + \varepsilon_t \quad (11)$$

where the factors are single-market proxies constructible from our data rather than the published cross-sectional factors: Carry is the average weekly spot return of the classic carry crosses AUD/JPY and NZD/JPY (a spot-only proxy for the Lustig et al., 2011 carry factor, omitting interest accrual), Mom is the return of a 12-week time-series momentum rule on EUR/JPY itself (in the spirit of Menkhoff et al., 2012), and Dollar is the DXY weekly return (proxying Verdelhan, 2018). Constructing the published factors requires a G10 forward-rate cross-section we do not use; the proxies capture the same directional exposures at the single-pair level, and we label them as proxies throughout. Because the strategy is out of the market most weeks, we estimate the regression twice: on the full weekly sample (where the dependent variable is structurally zero in out-of-position weeks) and on in-position weeks only, whose count is the effective sample size.

Table 9: Factor Attribution Regression (HAC-Corrected)

Variable	Full sample ($n = 504$)		In-position weeks ($n = 55$)	
	Coefficient	t -stat	Coefficient	t -stat
Intercept (α)	−0.00014	−0.55	−0.00145	−0.86
Carry (β_1)	0.0154	0.52	0.325	1.40
Momentum (β_2)	−0.0470	−2.09	−0.824	−3.53
Dollar (β_3)	0.0128	0.57	0.146	0.96
R^2	0.010		0.187	
F -statistic	2.06		6.14	

Note: Newey-West HAC standard errors with 4 lags. The full sample includes flat weeks; the in-position sample is the effective exposure sample. The intercept is not significant in either regression. The negative momentum loading is statistically significant in this specification, but the factors are single-market proxies and the in-position sample remains small.

No Evidence of Positive Residual Alpha. The intercept is negative and statistically indistinguishable from zero in both specifications ($t = -0.55$ full sample; $t = -0.86$ on in-position weeks). This is evidence against a positive residual-alpha claim, not proof that the population intercept is exactly zero.

Factor Loadings and Effective Sample. The corrected longer holds raise the effective sample to 55 weeks. The negative momentum loading is material in both regressions, while carry and dollar loadings are not significant. Because the factors are proxies and the exposed sample is small, this is a specification check rather than a definitive factor decomposition.

5.6 Weekly-Return Tail Distribution Analysis

A concern with any exceedance-based signal is whether extreme observations reflect a stable tail distribution or sparse outliers that may not recur out-of-sample. As a separate diagnostic, we characterize absolute Friday-close-to-Friday-close EUR/JPY returns using Extreme Value Theory (McNeil and Frey, 2000; Coles, 2001). This is not the same sample as tail alpha, which flags daily exceedances before

Friday sampling. The exercise therefore describes weekly return magnitudes and does not establish an upside-asymmetry story.

Exceedance Preprocessing. Financial tail events exhibit temporal clustering (volatility clustering), violating the GPD independence assumption. We apply runs declustering with minimum inter-exceedance separation of 5 weeks, following [Coles \(2001\)](#). Of the 26 raw exceedances above the 95th percentile threshold, declustering yields 20 cluster maxima used for GPD estimation. We estimate the extremal index $\theta = 0.83$ via the Ferro–Segers intervals estimator ([Ferro and Segers, 2003](#)), with a 95% confidence interval of $[0.60, 1.00]$ obtained by bootstrap resampling of the inter-exceedance times themselves ($B = 2000$)—a scheme that preserves the gap distribution the estimator is built on and, unlike a naive index bootstrap, produces an interval containing its own point estimate.

Methodology. We fit a Generalized Pareto Distribution (GPD) to cluster maxima above the 95th percentile threshold. The GPD is characterized by shape parameter ξ (determining tail heaviness) and scale parameter σ . Positive ξ indicates Pareto-type heavy tails; $\xi = 0$ corresponds to exponential decay; negative ξ implies bounded tails.

Table 10: GPD Parameter Estimates—Weekly Absolute Returns (Declustered)

Parameter	Estimate	95% CI
Shape (ξ)	−0.25	[−1.49, 0.27]
Scale (σ)	0.0089	—
Threshold (95th pctl)	0.0244	—
Extremal index (θ)	0.83	[0.60, 1.00]
Raw exceedances	26	—
Cluster maxima	20	—
KS test p -value	0.971	Fail to reject

Note: GPD fitted via maximum likelihood to cluster maxima of weekly absolute returns above the 95th percentile after runs declustering (5-week separation). ξ CI from a nonparametric bootstrap of cluster maxima ($B = 2000$); θ CI from bootstrap resampling of inter-exceedance times ($B = 2000$). KS = Kolmogorov-Smirnov goodness-of-fit test. Generated by `analysis/full_pipeline.py`; an earlier version of this table reported a θ confidence interval of $[0.42, 0.63]$ that excluded its own point estimate—an invalid interval produced by a biased index-resampling scheme, replaced here.

Tail Characterization. The estimated shape parameter $\hat{\xi} = -0.25$ (95% CI: $[-1.49, 0.27]$) is imprecise. The interval admits bounded, exponential-type, and moderately Pareto-type behavior; the data therefore do not support a strong tail-classification claim. The Kolmogorov-Smirnov test does not reject the fitted GPD ($p = 0.97$), but with only 20 cluster maxima it has limited diagnostic power.

Clustering Dynamics. The extremal index $\theta = 0.83$ indicates weak temporal clustering in tail events: the mean cluster size is $1/\theta \approx 1.2$, and the confidence interval reaches 1.0 (no clustering). Exceedances arrive close to independently, with mild GARCH-type persistence ([Bollerslev, 1986](#)) at most. Clustering therefore does little to reduce the effective sample size for tail inference; the binding constraint is simply the small number of exceedances.

Implications. EVT provides no affirmative evidence for the lottery-like right-tail mechanism required by the trading story. It does not justify assuming a particular tail class or finite variance outside this sample.

5.7 Multiple Testing Corrections

With five alpha types examined across multiple statistical tests, data-snooping concerns arise: apparent significance may reflect chance discoveries from multiple hypothesis testing. We address this through

both formal corrections and bootstrap-based reality checks.

Bonferroni Corrections. Table 2 applies plain Bonferroni correction at family size $m = 5$ (one skewness test per alpha type; $\alpha^* = 0.01$ for the 5% family level). At the corrected threshold, four of five alpha types reject *normality*, but the asymmetry inventory is far thinner: only coverage alpha’s skewness survives both the correction and, more demanding, the block-bootstrap dependence adjustment. The reported tests do not reject normality for fast alpha; this is a useful control result, not proof that its population distribution is Gaussian.

Bootstrap Implementation. White’s Reality Check and Hansen’s SPA employ the stationary block bootstrap of Politis and Romano (1994) with expected block length $\ell = 4$ weeks. The 13 candidates comprise five simplified asymmetry rules, the full rule, three momentum rules, two mean-reversion rules, always-long, and one seeded random candidate. Bootstrap replications: 1,000. The formal benchmark is zero weekly return.

White’s Reality Check. Following White (2000), we test the candidate universe against a zero-return benchmark. The Reality Check does not reject the null of no superior strategy ($RC = 0.02$, $p = 0.15$). The seeded random position sequence is one prespecified candidate, not the formal benchmark; it happens to have the highest realized sample mean.

Hansen’s SPA Test. The Superior Predictive Ability test (Hansen, 2005) agrees ($SPA = 1.90$, $p = 0.26$): no candidate demonstrates superior predictive ability over the zero-return benchmark at conventional levels.

Table 11: Data-Snooping Correction Results

Test	Statistic	p -value	Interpretation
White’s Reality Check	0.02	0.15	Not significant
Hansen’s SPA	1.90	0.26	Not significant
# Candidate Strategies	13	—	—

Note: Candidates: simplified asymmetry (5), full two-sided asymmetry (1), momentum (3), mean reversion (2), always-long (1), and a seeded random candidate (1). The formal benchmark is zero weekly return. All candidates use one execution lag. Stationary bootstrap: 1,000 replications, expected block length 4 weeks.

Interpretation. The tests find no statistically superior candidate in this prespecified universe. This is a failure to reject superiority, not proof that every possible strategy has zero expected return. The random candidate’s top sample mean underscores the instability of ranking noisy rules.

6 Conclusions

This paper investigated whether distributional asymmetries in FX alpha signals represent exploitable inefficiencies. Our findings are negative at every layer:

1. Only coverage alpha has a dependence-robust skewness interval that excludes zero; the other four intervals include zero
2. EVT tail-shape inference is too imprecise to establish Pareto-type heavy tails or exponential decay; clustering appears weak but is also estimated from few exceedances
3. The corrected asymmetry strategy loses 7.57% gross (15 holding episodes, Sharpe -0.17), with bootstrap intervals that include zero; walk-forward selection opens one episode in eight test years

4. Execution costs monotonically worsen the already negative gross result; omitted financing and end-of-bar execution remain limitations
5. Data-snooping tests find no statistically superior candidate against zero return, and exploratory cross-market outcomes are unstable

These null findings contribute to the literature in three ways. First, they demonstrate the gap between statistical and economic significance in quantitative trading research—a pattern likely more common than publication bias reveals (Harvey, 2017). Second, they document, from the inside, how easily higher-moment “stylized facts” can be manufactured by measurement choices: this paper’s own earlier version reported pronounced positive tail skewness that turned out to describe an unsigned magnitude series, right-skewed by construction. The corrected analysis is a case study in why skewness claims should be accompanied by the sign conventions, sampling frequency, and dependence-robust intervals that make them checkable. Third, they provide a documented negative result that may spare other researchers this particular dead end.

The methodological lesson is sharper than “backtests flatter”: most of the asymmetry was never there. Distributional anomalies in financial data are easily overstated at the measurement stage, before a single trade is simulated—and the sequence of checks (signed variables, dependence-robust intervals, training-data-only selection, data-snooping correction) dismantled this one layer by layer.

Future research might examine whether asymmetry exploitation is viable at higher frequencies (intraday) where both signal counts and cost structures scale differently, or in markets with different microstructure. Parallel findings in cryptocurrency markets—where an apparent infrastructure-versus-regulatory volatility asymmetry dissolves once cross-asset dependence and heavy tails are handled correctly (Farzulla, 2026b)—and in cross-market event studies more broadly (Farzulla, 2025) reinforce that distributional asymmetries are easily overstated and rarely translate into exploitable strategies. Our results echo that the burden of proof for asymmetry-based strategies should be high: an apparently sharp effect is insufficient evidence of economic value, and is often not robust evidence of a statistical effect either.

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Declarations

Conflict of Interest. The author declares no competing interests.

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Data Availability. The pipeline retrieves Yahoo Finance series or reads local cached CSV files. The analysis manifest records retrieval dates, row counts, software versions, and SHA-256 hashes. Raw snapshots are not committed because redistribution terms may apply. The original snapshots underlying the previously committed results were unavailable; exact reproduction requires those files from the owner.

Code Availability. Full replication code available at <https://github.com/dissensus-ai/alpha-asymmetry> under MIT License.

AI Assistance. Claude (Anthropic) assisted with manuscript drafting, code review, and statistical exposition. All research design, data analysis, and interpretation are solely the author's.

Author Contributions. Sole author.

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A Enlarged Figures

For improved readability, this appendix reproduces all figures at enlarged scale.

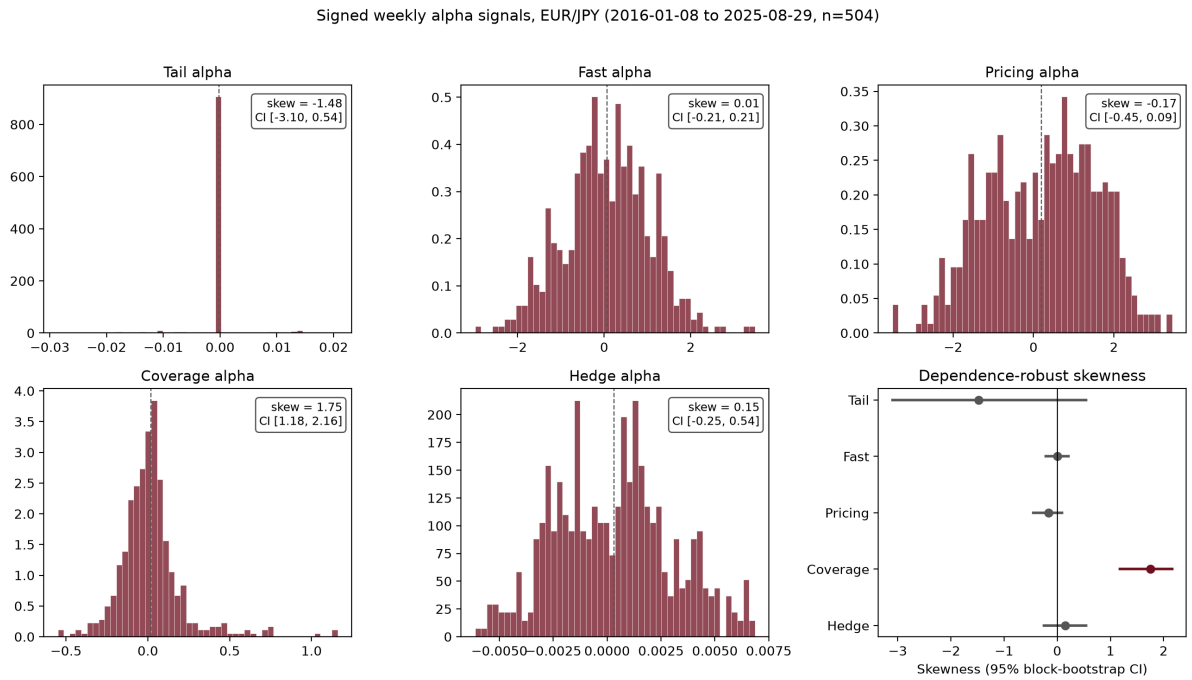


Figure 3: Alpha Asymmetry Analysis (enlarged). Distributional properties of five alpha types in the EUR/JPY analysis-ready sample (January 2016–August 2025).

Strategy equity curves and drawdowns, one-lag Friday-close convention

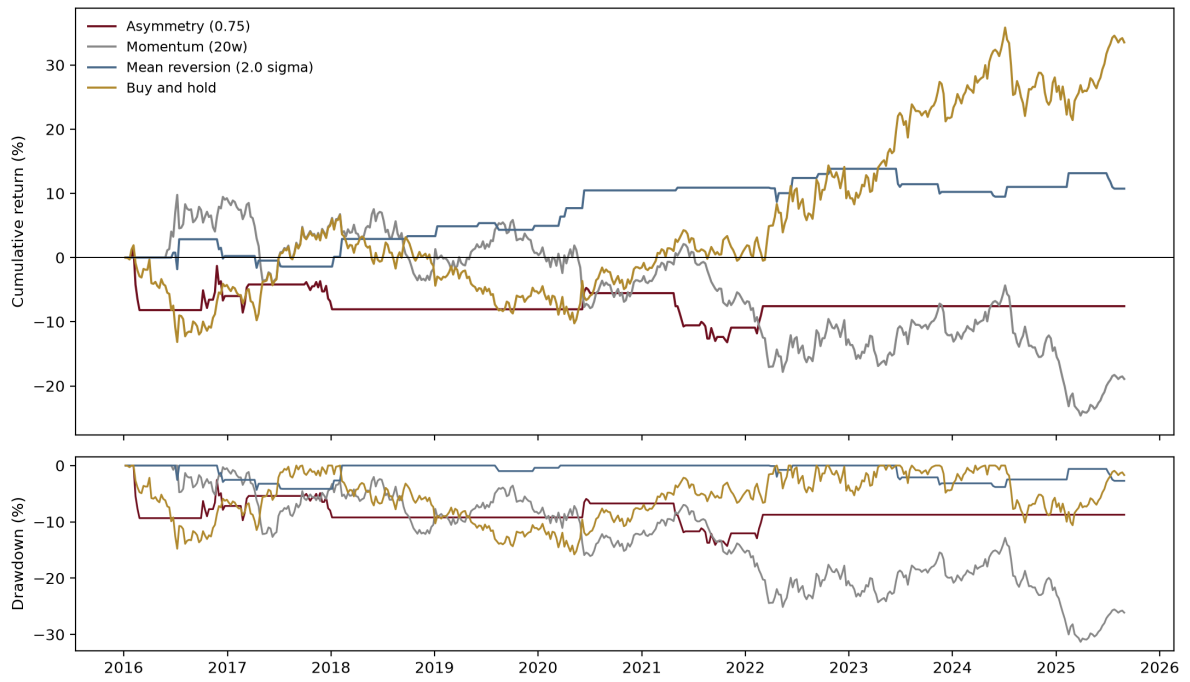


Figure 4: Strategy equity curves and drawdowns (enlarged). Cumulative gross returns and drawdown profiles for the asymmetry strategy, the 20-week momentum and 2.0σ mean-reversion benchmarks, and buy-and-hold on the 504-week EUR/JPY sample.